

# MATHMANIA

## Official Competition Rulebook

Mathmania is a multi-round mathematics competition designed to test speed, accuracy, logical reasoning, and depth of mathematical understanding. This rulebook outlines the format and rules for all participants.

The questions and activities across MathMania may draw from a variety of mathematical areas, including but not limited to:

- Arithmetic and number sense
- Estimation, ratios and proportions
- Algebraic reasoning
- Logical thinking and pattern recognition
- Basic geometry and spatial reasoning
- Graphs, tables, and data interpretation
- Error Detection
- Mathematical applications in real-life scenarios

Participants are not required to have advanced or university-level mathematics knowledge. However, clarity of concepts, logical and mathematical approach, and time management will be essential throughout the module.

## Competition Structure

Round 1: Kahoot (MCQ-based)

Round 2: Descriptive Round (Written problem-solving)

Round 3: Buzzer Round (Team-based final)

## ROUND 1: KAHOOT

Participants will compete in teams, with each team participating as one user on the digital platform Kahoot. Team members are expected to remain together during the round and coordinate their responses through discussion within the team only. The questions will have a fixed time limit and are designed with varying levels of difficulty throughout the round.

**Format:** Team (2 – 3 participants)

**Time:** 45 Minutes

**Rules:**

- All teams are required to join the Kahoot session using the assigned team name as instructed.
- Once an answer is submitted or the time expires, it cannot be changed.
- Only one response per team will be recorded.
- Use of multiple devices, switching team members, or attempting to join the quiz more than once is strictly prohibited.

## **ROUND 2: DESCRIPTIVE ROUND**

Participants will be presented with a mathematical situation that progressively evolves over time, requiring to adapt their approach as new information or conditions are introduced. Responses will be completed in written form within strict time constraints, and solutions must be presented according to the instructions provided.

**Format:** Individual or Team (2–3 participants)

**Time:** 45 minutes

**Rules:**

- All responses must be written with clear working and submitted within the allocated time.
- Teams must work only with the information provided during the round.
- No external resources or electronic devices are permitted unless explicitly allowed.
- Partial credit will be awarded for correct reasoning but incomplete answer.

## **ROUND 3: BUZZER ROUND**

Round 3 will be conducted as a buzzer-based round, where teams will compete by responding to questions in real time. Questions will be presented one at a time, and teams must buzz in to earn the opportunity to answer.

**Format:** Head-to-head teams (2–3 participants per team)

**Rules:**

- Only the team that buzzes in first may attempt to answer the question.
- Teams must wait for the question to be fully read before buzzing.
- Answers must be given immediately once the team has buzzed in; no additional time will be granted.
- Teams are not allowed to use any external resources or electronic devices unless explicitly permitted.
- Each correct answer will earn the team points, while incorrect answers will receive no points.
- If a team buzzes but fails to provide an answer, negative points will be applied, and the question will be passed to the next eligible team.
- Communication with other teams is strictly prohibited.
- Any form of unfair play, including deliberate disruption or interference, will lead to disqualification.

## **General Guidelines & Rules**

To ensure a fair, respectful, and smooth competition, all participants are expected to adhere to the following guidelines throughout the MathMania module:

- Participants must work independently or within their assigned teams, as applicable, and must not share answers or assist other teams.
- Cheating, plagiarism, or use of unauthorized resources will result in immediate disqualification.
- All work, screens, and submissions must be kept private.
- Respect towards judges, organisers, and fellow participants is mandatory. Disruptive or inappropriate behaviour will not be tolerated.
- Instructions provided by organisers and judges must be followed at all times; failure to comply may lead to penalties or disqualification.
- Electronic devices, calculators, or other reference materials are not allowed unless explicitly permitted.
- Participants are expected to be punctual and prepared for all rounds.
- Decisions made by judges and organisers are final and binding.